

HW 09 SOLUTIONS

Practice Problems

4.123a

$Y \sim \text{Beta}(\alpha = 4, \beta = 3)$, $k = \frac{\Gamma(4+3)}{\Gamma(4)\Gamma(3)} = 60$
95th percentile: $\phi_{.95} = 0.84684$

4.127

For $\alpha = \beta = 1$, $f(y) = \frac{\Gamma(2)}{\Gamma(1)\Gamma(1)}y^{1-1}(1-y)^{1-1} = 1 \Rightarrow Y \sim \text{Uniform}(0, 1)$

4.165

The pdf of Y is in the form of a gamma density with $Y \sim \text{Gamma}(\alpha = 2, \beta = 0.5)$

(a) $c = \frac{1}{\Gamma(2)0.5^2} = 4$

(b) $E(Y) = \alpha\beta = 2(0.5) = 1$, $V(Y) = \alpha\beta^2 = 2(.5)^2 = 0.5$

(c) MGF: $M(t) = (1 - \beta t)^{-\alpha} = (1 - 0.5t)^{-2}$, $t < 2$

Submitted problems

4.126

$$(a) \int_0^y (6t - 6t^2)dt = 3y^2 - 2y^3, \text{ so } F(y) = \begin{cases} 0, & y < 0 \\ 3y^2 - 2y^3, & 0 \leq y \leq 1 \\ 1, & y > 1 \end{cases}$$

(b) not shown

$$(c) P(0.5 \leq Y \leq 0.8) = F(0.8) - F(0.5) = 1.92 - 1.092 - 0.75 + 0.25 = 0.396$$

4.130

$$\sigma^2 = E(Y^2) - \mu^2,$$

$$E(Y^2) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 y^{\alpha+1}(1-y)^{\beta-1} dy = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \frac{\Gamma(\alpha + 2)\Gamma(\beta)}{\Gamma(\alpha + 2 + \beta)} = \frac{(\alpha + 1)\alpha}{(\alpha + \beta)(\alpha + \beta + 1)}$$

$$V(Y) = \frac{(\alpha+1)\alpha}{(\alpha+\beta)(\alpha+\beta+1)} - \left(\frac{\alpha}{\alpha+\beta}\right)^2 = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$